

PROTRANSIT

Connecting Cities. Moving Goods Smarter.

- A public-transit-powered urban logistics coordination platform.
- Connecting Senders -> Pickup Hubs -> Public Transit (Bus/Metro/Rail) -> Destination Hubs -> Delivery Partners -> Receivers.
- Preserves strict chain of custody via Pickup QR and Receiver OTP.
- Eliminates duplicate long-distance road trips by utilizing existing permitted transit capacity.

Problem Statement & Urban Context

Cities are moving more. Roads are carrying more.

- Urban Road Congestion: Skyrocketing on-demand parcel deliveries exacerbate peak-hour city traffic.
- Unused Public Transit Capacity: Buses, metros, and suburban trains run off-peak with empty baggage/parcel storage capacity.
- Escalating Last-Mile Costs: Direct point-to-point private couriers incur high operational overheads and duplicate infrastructure.
- Air Pollution & Carbon Emissions: Dedicated two-wheeler and van parcel runs contribute significantly to vehicular emissions.

The ProTransit Solution

Turn existing mobility infrastructure into a logistics backbone.

- Multimodal Transit Backbone: Coordinates permitted parcel movement across scheduled bus and metro networks.
- Designated Transit Hubs: Utilizes existing bus stands and metro stations as secure handover and sorting points.
- Controlled Chain of Custody: High-security QR code check-in at hubs with encrypted receiver OTP verification.
- Verified Last-Mile Partners: Independent delivery riders perform first-mile and last-mile pickup and delivery.

Multimodal Routing & Price Advantage

Demonstrated 47% Cost Savings & Lower Emissions

- ProTransit Multimodal: INR 45 | 2h 20m | Bus -> Metro -> Hub -> Last-Mile Partner | 3.2 kg CO2 Saved.
- Standard Private Courier: INR 85 | 1h 45m | Direct point-to-point courier road trip.
- Local Express Delivery: INR 95 | 1h 35m | Dedicated two-wheeler rider trip across city.
- Shared Infrastructure Advantage: By piggybacking on scheduled public transit, marginal cost per parcel is drastically lower.

Graph Routing Engine & Architecture

Deterministic Network Scheduling Model

- Mathematical Graph Model $G = (V, E)$: Vertices V = Transit Hubs; Edges E = Transit schedules (Bus/Metro/Rail).
- Optimization Function: $\text{Min}(\text{Cost}, \text{Transfer Delay}, \text{Carbon Impact})$ subject to parcel dimension and weight limits.
- Algorithmic Implementation: Time-Dependent Shortest Path (TDSP) based on Dijkstra and A* variations.
- Scalable Future System: Microservices on FastAPI, PostgreSQL, WebSockets for live telemetry, and Cloudflare CDN.

Chain-of-Custody & Trust Model

Every Handover Leaves an Immutable Trace

- Unique Shipment ID: Unique parcel tracking identifier (e.g. TX10231).
- Pickup QR Code: Scanned by hub operator at pickup hub to initiate transit journey.
- Real-time Transit Timeline: Tracks parcel arrival, metro transfers, and dispatch to last-mile partner.
- Masked Communication: Customer and rider contact details are completely protected through masked routing.
- Receiver OTP Verification: Parcel handed over only upon entering the secret 4-digit OTP (e.g. 1234).

Prototype Status & SIH 2026 Deliverables

Fully Verified Android App & Web Presence

- Android Release APK: 48.13 MB, tested on Android API 34 with 0 layout overflows and 0 analyzer issues.
- Tested End-to-End Flow: 14-screen flow verified from Home to Send Parcel, Price Comparison, QR, Tracking, and OTP.
- Responsive Web Companion: Modern React + TypeScript + Tailwind website with interactive hub graph & browser demo.
- Cloudflare Deployment: Deployment-ready for global edge distribution via Cloudflare Pages.

Project Roadmap & Scaling

From SIH Prototype to City-Wide Deployment

- Phase 1 (Prototype): Flutter Android mobile app, simulated routing engine, mock data, and documentation.
- Phase 2 (Pilot Corridor): Partnership with municipal transport corporation for single pilot route with smart lockers.
- Phase 3 (Production Scale): Real-time GTFS transit integration, partner app, UPI escrow, and automated sorting.
- Phase 4 (City Intelligence): AI-powered capacity forecasting, dynamic route pricing, and statewide network expansion.